

# Web Application Testing

Types of Tests • What to Test • How to Test It

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# 1. Web Testing Overview

Web application testing validates that the application behaves correctly, performs well, is secure, and works consistently across browsers, devices, and network conditions. A comprehensive web test strategy covers all of the following disciplines.

## 2. Functional Testing

Verify the application does what the requirements say it should do.

| What to Test                                        | Techniques                                                   | Tools                                         |
|-----------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------|
| All user-facing features vs acceptance criteria     | Scripted test cases per User Story; exploratory charters     | Azure DevOps Test Plans, Playwright, Selenium |
| Form validation — required fields, formats, lengths | Boundary value analysis; equivalence partitioning            | Manual, Playwright form-fill scripts          |
| Navigation — all links, menus, breadcrumbs          | Link-crawl; manual navigation testing                        | Playwright, Screaming Frog (link checker)     |
| Search & filtering — results accuracy and relevance | Known-data test sets; zero-result cases; filter combinations | Manual, Postman (API layer), Playwright       |
| CRUD operations — Create, Read, Update, Delete      | End-to-end with DB verification                              | Manual + SQL validation, REST Assured         |
| Email & notification workflows                      | Trigger events; verify delivery, content, links              | Mailtrap, Mailinator, manual                  |
| File upload & download                              | Valid/invalid file types; max size; download integrity       | Manual, Playwright file API                   |
| Error pages — 404, 500, 403                         | Navigate to non-existent routes; trigger server errors       | Manual, curl                                  |

## 3. UI & Visual Testing

| What to Test             | Specific Checks                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| Layout integrity         | No overlapping elements; correct alignment; responsive breakpoints (320px, 768px, 1024px, 1440px)        |
| Typography               | Correct fonts, sizes, weights, colours per design system — no fallback fonts rendered                    |
| Images                   | All images load; correct alt text present; no broken image icons; correct aspect ratio                   |
| Colour & contrast        | WCAG AA minimum contrast ratio 4.5:1 for body text; 3:1 for large text                                   |
| Animations & transitions | Smooth at 60fps; no janky repaints; reduced-motion OS setting respected                                  |
| Empty states             | Empty search results, empty cart, new user dashboard — all have designed empty states, not blank screens |
| Loading states           | Skeleton loaders or spinners present for all async operations; no blank flashes                          |
| Visual regression        | Screenshot diff vs baseline on every regression run — catch unintended CSS changes                       |

## 4. Cross-Browser Testing

| Browser           | Engine | Priority                            | Tool                                      |
|-------------------|--------|-------------------------------------|-------------------------------------------|
| Chrome (latest)   | Blink  | P1 — always                         | Playwright Chromium, Selenium             |
| Firefox (latest)  | Gecko  | P1 — always                         | Playwright Firefox, Selenium              |
| Safari (latest)   | WebKit | P1 — if macOS/iOS users in audience | Playwright WebKit, BrowserStack           |
| Edge (latest)     | Blink  | P2 — enterprise apps                | Playwright Chromium (same engine)         |
| Chrome on Android | Blink  | P1 — if mobile traffic significant  | BrowserStack, Playwright mobile emulation |
| Safari on iOS     | WebKit | P1 — if iOS users in audience       | BrowserStack real device                  |
| Samsung Internet  | Blink  | P3 — Android-heavy audiences        | BrowserStack                              |

**Minimum cross-browser matrix:** Chrome, Firefox, and Safari (WebKit) on desktop + Chrome on Android + Safari on iOS. Run Playwright's 3 built-in browser engines for desktop; use BrowserStack for real iOS device testing.

## 5. API Testing

| Test Type        | What to Check                                                                                      | Tools                                                                       |
|------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Contract testing | Response schema matches API spec (Swagger/OpenAPI); all fields present, correct types              | Postman schema validation, REST Assured                                     |
| Functional       | Correct data returned for known inputs; CRUD operations behave correctly                           | Postman, REST Assured, PyTest                                               |
| Authentication   | Valid token → 200; no token → 401; expired token → 401; wrong scope → 403                          | Postman, curl                                                               |
| Error responses  | 400 for malformed request; 404 for missing resource; 422 for validation failure; 500 never exposed | Postman negative collections                                                |
| Data integrity   | API response matches DB record; no data transformation errors                                      | Postman + SQL validation                                                    |
| Performance      | Response time under SLA (e.g. 800ms for read endpoints)                                            | Postman Tests tab:<br><code>pm.expect(responseTime).to.be.below(800)</code> |
| Rate limiting    | Exceed rate limit → 429 response with Retry-After header                                           | Postman Collection Runner with iteration count                              |

## 6. Performance Testing

| Test Type                   | Goal                                                                            | Tool                                                  |
|-----------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|
| Page load — Core Web Vitals | LCP < 2.5s; FID/INP < 100ms; CLS < 0.1                                          | Lighthouse (Chrome DevTools), PageSpeed Insights      |
| Network throttling          | App usable on 3G (1.5 Mbps); no layout shift on slow load                       | Chrome DevTools throttle presets                      |
| Load testing                | System handles concurrent user target (e.g. 500 users)                          | k6, JMeter, Azure Load Testing                        |
| Stress testing              | Find breaking point — at what concurrency does error rate spike?                | k6, JMeter                                            |
| API response times          | All GET endpoints < 800ms; POST endpoints < 2s under normal load                | Postman, REST Assured timing assertions               |
| Memory leaks                | Browser memory stable after 30 min of normal usage; no steadily increasing heap | Chrome DevTools Memory tab — heap snapshot comparison |

## 7. Security Testing

| Attack Vector              | What to Test                          | How                                                                               |
|----------------------------|---------------------------------------|-----------------------------------------------------------------------------------|
| SQL Injection              | Text inputs accepting SQL fragments   | Enter: ' OR 1=1-- in search/login fields; verify input sanitised                  |
| XSS (Cross-Site Scripting) | Script injection in input fields      | Enter: alert('xss'); verify escaped, not executed                                 |
| CSRF                       | State-changing requests without token | Check all POST/PUT/DELETE requests include CSRF token; replay without token → 403 |
| Authentication bypass      | Access protected pages without login  | Navigate directly to /dashboard, /admin — verify 401/redirect to login            |
| Sensitive data exposure    | Passwords, tokens in URLs or logs     | Check URL bar, browser history, server logs for credentials                       |
| Broken access control      | User A accessing User B's data        | Swap resource IDs in API calls; verify 403 for unauthorised resources             |
| Session management         | Session fixation, overly long TTL     | Check cookie flags: HttpOnly, Secure, SameSite; test session expiry               |
| Password policy            | Weak passwords accepted               | Attempt: '123456', 'password', single-char — verify rejection                     |

## 8. Accessibility Testing (WCAG 2.1 AA)

| Test                        | Requirement                                                                  | Tool                                              |
|-----------------------------|------------------------------------------------------------------------------|---------------------------------------------------|
| Keyboard navigation         | All interactive elements reachable and operable via Tab/Enter/Arrow keys     | Manual keyboard-only session                      |
| Screen reader compatibility | All images have alt text; form fields have labels; headings are hierarchical | NVDA + Firefox; JAWS + Chrome; VoiceOver + Safari |
| Colour contrast             | Body text ≥ 4.5:1 ratio; large text ≥ 3:1                                    | axe DevTools, Colour Contrast Analyser            |

| Test                | Requirement                                                                                             | Tool                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Focus indicators    | Visible focus ring on all interactive elements — never <code>display:none</code> on <code>:focus</code> | Manual + axe scan                                              |
| Form error messages | Errors identified by text, not colour alone; linked to field via <code>aria-describedby</code>          | Manual + axe                                                   |
| Automated scan      | Catch ~35% of WCAG violations automatically                                                             | axe DevTools browser extension; Lighthouse Accessibility score |
| Reduced motion      | Animations stop or slow when OS 'Reduce Motion' is enabled                                              | macOS/Windows accessibility settings + manual check            |